

**NATIONAL UNIVERSITY OF SINGAPORE****CS4268 – QUANTUM COMPUTING**

(Semester 2: AY2021/22)

Time Allowed : 2 Hours

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**INSTRUCTIONS TO STUDENTS**

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FOUR** questions for a total of **40** marks, and comprises **THREE** printed pages (excluding this page).
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is an OPEN BOOK assessment.
6. Permitted devices: electronic calculators

The following information is useful for all questions.

1. The Pauli matrices are given by

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

2. Let  $N$  be an integer greater than 1, and define the following sets:

$$\mathbb{Z}_N = \{0, \dots, N-1\} \quad \mathbb{Z}_N^* = \{a \in \mathbb{Z}_N \mid \gcd(a, N) = 1\}.$$

3.  $\forall n \geq 2: \sum_{j=0}^{n-1} \omega_n^j = 0$ , where  $\omega_n = e^{\frac{2\pi i}{n}}$ .

4. For  $d \geq 2$ , the Quantum Fourier Transform matrix  $QFT_d$  is defined as follows:

$$QFT_d = \frac{1}{\sqrt{d}} \sum_{x=0}^{d-1} \sum_{y=0}^{d-1} \omega_d^{xy} |y\rangle \langle x|.$$

5. For a unitary  $V$ , the circuit in Figure 1 represents control- $V$  gate.

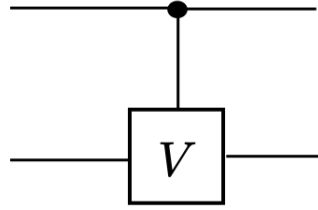


Figure 1

6. For all gates we specify action on basis states and action on superposition states is determined by linearity.

**Question 1. (10 marks)** [Completely new]

Calculate probability of getting  $|00\rangle$  and  $|10\rangle$  as output when  $|\phi\rangle = \sqrt{\frac{1}{2}}|00\rangle + \sqrt{\frac{1}{2}}|11\rangle$  is given as the input to the circuit in Figure 2 (consider the qubits 1 and 2 from top to bottom).

**Note:** In the given circuit,  $H$  denotes the Hadamard gate,  $Z$  denotes the phase gate and  $\mathcal{M}$  denotes measurement.

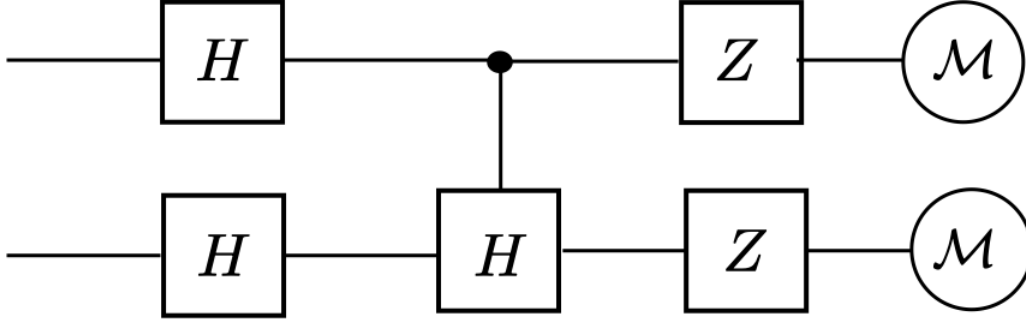


Figure 2

**Question 2. (5+5 marks)** [Completely new]

- Construct a classical circuit using AND, OR, NOT gates and fanout, that takes three bits  $a, b, c$  as input and outputs  $f(a, b, c) = c \oplus (a \wedge b) \oplus (a \vee b)$ .
- Construct a quantum circuit  $U$  acting on 3-qubits using only CNOT, Toffoli gates and not using any additional ancillia such that,

$$U|a\rangle|b\rangle|c\rangle = |a\rangle|b\rangle|c \oplus (a \wedge b) \oplus (a \vee b)\rangle$$

for every  $a, b, c \in \{0, 1\}$ .

**Question 3. (5+5+5 marks) [Completely new]**

Let  $S = \{x : w(x) \bmod 2 = 0\}$ , where  $w(x)$  is the Hamming weight of the string  $x \in \{0, 1\}^n$ . Let  $f$  be a function from  $\{0, 1\}^n$  to  $\{0, 1\}$  such that  $f(x)$  takes value 1 for exactly  $1/4$  fraction of set  $S$  and 0 elsewhere. Let

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{x \in S} |x\rangle.$$

- Construct a unitary  $U_S$  such that  $U_S|0^n\rangle = |S\rangle$ .
- Assume that you can apply circuits for  $U_S$ ,  $U_S^\dagger$  and  $R_0 = 2|0^n\rangle\langle 0^n| - \mathbb{1}$  unitaries. Construct the unitary  $R_S = 2|S\rangle\langle S| - \mathbb{1}$ , using  $U_S$ ,  $U_S^\dagger$  and  $R_0$  unitaries. Here  $\mathbb{1}$  represents the identity matrix.
- Recall that phase oracle for a function  $f$  denoted as  $Z_f$  is defined as  $Z_f|x\rangle = (-1)^{f(x)}|x\rangle$ . Using  $Z_f$ , starting from state  $|0^n\rangle$ , describe an algorithm to output an  $x_0$  such that  $f(x_0) = 1$  with probability 1. You need not draw complete circuit, only unambiguous and clear description is enough. How many times do you need to use  $Z_f$ ?

**Question 4. (5 marks) [Completely new]**

Let  $p$  be a prime number and  $b \in \mathbb{Z}_p^*$ . For any  $a \in \mathbb{Z}_p^*$ , assume you have access to a unitary  $V_a$  such that

$$V_a|i\rangle = |ai \bmod p\rangle \quad \text{for all } i \in \mathbb{Z}_p.$$

Also, assume that you have access to the unitary  $QFT_p$ . Construct a unitary  $U$  such that

$$U|i\rangle = \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \omega_p^{bik} |k\rangle \quad \text{for all } i \in \mathbb{Z}_p.$$

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**END OF PAPER**